Local no-slip Navier–Stokes equations under stochastic parabolicity

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Abstract

We consider the three-dimensional incompressible Navier–Stokes equation on a bounded smooth domain with no-slip boundary condition and finitely many smooth, divergence-free, boundary-tangent transport noises. We prove local pathwise well-posedness under the stochastic parabolicity condition $\nu I - \frac{1}{2}\sum_k b_k \otimes b_k \geq \delta I > 0$, without an additional smallness assumption on the noise. For smooth compatible initial data the solution is continuous in $B^{1/2}_{4,4}$ and belongs locally to L^4 in time with values in $W^{1,4}_0$. Uniqueness holds in this entire path class, and the solutions depend continuously in probability on the initial data after common stopping. The proof first controls constant transport on periodic channels and full interior tori. Localization through the Stokes energy form preserves the weak Dirichlet scale; a quantitative square-root comparison makes its errors small enough to recover the physical-domain estimate.

1 Introduction

Consider the Itô equation

$$du = [\nu \mathbb{P}_D \Delta u - \mathbb{P}_D \operatorname{div}(u \otimes u)]\,dt + \sum_{k=1}^{K} \mathbb{P}_D[(b_k \cdot \nabla)u]\,dW_t^k, \qquad \operatorname{div} u = 0, \quad u|_{\partial D} = 0. \tag{1.1}$$

Here $D \subset \mathbb{R}^3$ is bounded and smooth, $\nu > 0$ is the viscosity, $\mathbb{P}_D$ is the Helmholtz projection, and the finite family b_k is smooth, solenoidal, and tangent to the wall. Transport noise acts at the same differential order as the square root of viscosity. Its quadratic variation therefore changes the coercive second-order part of an energy estimate. The natural positivity condition is

$$\nu I - \tfrac{1}{2}\sum_k b_k(x) \otimes b_k(x) \geq \delta I > 0 \qquad (x \in \overline{D}).$$

The question is whether this positivity suffices for a strong local theory with no-slip walls, even when the transport cannot be treated as a small perturbation.

The deterministic weak Stokes scale places convection in a negative Sobolev space and reduces local existence to an estimate for the linear equation. Prüss and Wilke [10, Section 5] develop this formulation for deterministic Navier–Stokes equations. Stochastic maximal regularity gives an analogous route once the linear noise has been controlled; see Agresti and Veraar [1]. Their survey distinguishes the whole-space non-small transport theory from the no-slip boundary question, which appears as Open Problem 9. In the presence of walls, pressure and the zero vector trace interact with localization. In particular, Helmholtz projection enforces the normal constraint but does not in general preserve the tangential zero trace.

Boundary results also depend on the precise equation and solution class. Goodair's strong two-dimensional theory uses Navier boundary conditions [7]; its discussion explains the difficulty of the corresponding no-slip estimates. His no-slip channel inviscid-limit result [8] uses weak martingale solutions and a transport-stretching model. These are different questions from pathwise local well-posedness for (1.1) on arbitrary smooth three-dimensional domains. We address the finite smooth-coefficient formulation in AIM Problem 458 [2]. This is a specified part of the broader no-slip problem in [1].

Let

$$\mathcal{E} = \{h \in B^{1/2}_{4,4}(D;\mathbb{R}^3) : \operatorname{div} h = 0,\ \operatorname{Tr} h = 0\}, \qquad V_4 = W^{1,4}_{0,\sigma}(D), \qquad Y_4 = (W^{1,4/3}_{0,\sigma}(D))^*.$$

The trace in $\mathcal{E}$ is well defined since $1/2 > 1/4$. Equation (1.1) is understood as a weak solenoidal integral identity in Y_4. In particular convection is paired against $\varphi \in W^{1,4/3}_{0,\sigma}$ by $\int_D (u \otimes u) : \nabla\varphi$.

Theorem 1.1. *Let $D \subset \mathbb{R}^3$ be any bounded smooth domain and $\nu > 0$. Let $b_1, \dots, b_K \in C^\infty(\overline{D};\mathbb{R}^3)$ be divergence-free, tangent to ∂D, and satisfy the displayed positivity condition for some $\delta > 0$. On any filtered probability space satisfying the usual conditions, with independent standard Brownian motions $W^1, \dots, W^K$, every deterministic smooth solenoidal datum h with zero vector trace admits an almost surely positive stopping time τ and an adapted solution of (1.1), $u(0) = h$, with paths in*

$$C([0,\tau]; B^{1/2}_{4,4}(D;\mathbb{R}^3)) \cap L^4(0,\tau; W^{1,4}_0(D;\mathbb{R}^3)). \tag{1.2}$$

Any two adapted solutions in this entire class with the same data and Brownian motions agree up to their common lifetime.

Fixing the domain, coefficients, and Brownian coupling, let smooth admissible $h_n \to h$ in $\mathcal{E}$. The corresponding solutions can be chosen with a positive base time τ and positive pairwise common stops $\sigma_n \le \tau$ such that, for each $\epsilon > 0$,

$$\mathbb{P}\left\{ \sup_{t \le \sigma_n} \|u_n(t) - u(t)\|_{\mathcal{E}} + \left(\int_0^{\sigma_n} \|u_n - u\|^4_{V_4}\, dt \right)^{1/4} > \epsilon \right\} \longrightarrow 0, \tag{1.3}$$

$$\mathbb{P}\{\sigma_n < \tau\} \longrightarrow 0.$$

Thus perturbations reach the same base interval with probability tending to one, and converge in the full asserted path topology on their common intervals. The theorem imposes no amplitude smallness beyond stochastic parabolicity and no planar-span restriction on the interior noise. It is local in time; it makes no assertion of global three-dimensional regularity. The proof uses smooth finite noise families and does not establish the broader rough or infinite-noise formulations of the no-slip question.

The main work is linear. In a flat channel, constant tangential transport is removed by random translation, leaving an anisotropic Stokes operator. Its pressure and heat roots can coincide even at a positive resolvent parameter. Divided differences keep the resolvent regular there. The matrix imposing the two wall conditions has a uniformly bounded inverse, also after weighting the normal component. These two bounds preserve the smaller normal trace of the forcing and the factor k/a in the slow tangential response, where k and a are the pressure and heat frequency scales. They give the required operator-valued multiplier bounds. Interior points instead use a full three-dimensional torus, so their frozen vectors may have arbitrary rank.

To pass to a curved wall, we localize the Stokes energy form between the physical and flat domains. This preserves both the no-slip positive space and its weak dual. Comparing it with Piola transport requires control of the nonorthogonal pressure term. Negative-order pressure estimates

and the perturbation theorem of Kalton, Kunstmann, and Weis [9] give a quantitative square-root comparison. Its smallness survives summation over the cover; compact errors reduce to the physical L^2 energy estimate. A finite continuation in the noise amplitude then constructs the linear solution operator throughout the parabolicity range. An adapted state cutoff gives the nonlinear solution, while a separately stopped energy argument proves uniqueness without moment assumptions on the competing paths.

Section 2 fixes the weak formulation and states the linear estimate. Section 3 proves the flat estimates; Sections 4 and 5 localize and construct the physical linear solution. Section 6 proves the nonlinear assertions. Throughout, Fourier modes are $e^{i\xi\cdot x}$, positive Stokes operators have sign $-\mathbb{P}\Delta$, and complex square roots have positive real part. Constants may depend on the fixed domain and coefficients and need not stay bounded as $\delta \downarrow 0$.

2 Weak Stokes formulation and the linear estimate

Let $D \subset \mathbb{R}^3$ be bounded with smooth boundary. We use the ordinary Helmholtz projection $\mathbb{P}_D$ and set, for $1 < q < \infty$,

$$H_q = L^q_\sigma(D), \qquad V_q = W^{1,q}_{0,\sigma}(D), \qquad Y_q = (V_{q'})^*, \qquad q' = q/(q-1).$$

Here H_q consists of solenoidal fields with zero weak normal trace; V_q has zero vector trace. Duals use the integral pairing. Let $A_D = -\mathbb{P}_D\Delta$ be the positive strong Dirichlet Stokes operator, with domain $W^{2,q}(D;\mathbb{R}^3) \cap V_q$. Its weak realization, denoted $A_{D,w}$, is the form

$$\langle A_{D,w}u, v\rangle = \int_D \nabla u : \nabla v, \qquad u \in V_q, \quad v \in V_{q'}. \tag{2.1}$$

For the equation in this paper the trace space is

$$\mathcal{E} = (Y_4, V_4)_{3/4,4} = \{h \in B^{1/2}_{4,4}(D;\mathbb{R}^3) : \operatorname{div} h = 0,\ \operatorname{Tr} h = 0\}. \tag{2.2}$$

The trace on the right is the Besov trace. We write $[X_0, X_1]_s$ for complex interpolation and $(X_0, X_1)_{s,p}$ for real interpolation. All spatial norms are inhomogeneous. The weak Stokes identifications in [10, Section 5, Proposition 5.1] give (2.2) and $[Y_q, V_q]_{1/2} = H_q$. They follow from the isotropic Stokes square-root scale; they will not be assumed for an anisotropic operator. The strong isotropic operator has a bounded H^∞ calculus on H_q by [9, Theorem 9.17]. The negative-half identification

$$J_q = (1 + A_D)^{-1/2} : Y_q \longrightarrow H_q$$

is an isomorphism, with $J_q(V_q) = \mathrm{D}(A_D)$ and $J_q A_{D,w} J_q^{-1} = A_D$. Thus the weak operator is similar to the strong one. We also use the resulting fractional-domain and interpolation identities for $1 + A_D$.

Fix a filtered probability space satisfying the usual conditions and independent real Brownian motions $W^1, \dots, W^K$. For predictable processes write

$$\mathcal{M}_T = C([0,T];\mathcal{E}) \cap L^4(0,T;V_4), \qquad \mathcal{X}_T = L^4(\Omega;\mathcal{M}_T),$$

with sum norms. The noise norm $\gamma(\mathbb{R}^K, H_4)$ is equivalent to $\big\|(\sum_k |g_k|^2)^{1/2}\big\|_{L^4}$, with constants independent of finite-dimensional Hilbert indices. No boundary trace is imposed on g. Stochastic integrals below have their usual UMD-space meaning; the integral estimate used is [1, Proposition 2.4].

Let $b_k \in C^\infty(\overline{D};\mathbb{R}^3)$ be solenoidal and tangent to ∂D, and suppose

$$Q(x) := \nu I - \tfrac{1}{2}\sum_{k=1}^{K} b_k(x)\otimes b_k(x) \;\geq \delta I > 0, \qquad \nu > 0. \tag{2.3}$$

Set $B_k u = \mathbb{P}_D[(b_k\cdot\nabla)u]$. This is bounded $V_4 \to H_4$. The main analytic estimate is the following.

Theorem 2.1. *For each fixed $T_0 < \infty$, every $0 < T \leq T_0$, every $\mathcal{F}_0$-measurable $h \in L^4(\Omega;\mathcal{E})$, and predictable*

$$f \in L^4(\Omega\times(0,T);Y_4), \qquad g \in L^4(\Omega\times(0,T);\gamma(\mathbb{R}^K,H_4)),$$

there is a unique predictable $u \in \mathcal{X}_T$ solving

$$du + \nu A_{D,w}u\,dt = f\,dt + \sum_k (B_k u + g_k)\,dW^k, \qquad u(0) = h, \tag{2.4}$$

as an integral identity in Y_4. It satisfies

$$\|u\|_{\mathcal{X}_T} \leq C\left(\|h\|_{L^4(\Omega;\mathcal{E})} + \|f\|_{L^4(\Omega\times(0,T);Y_4)} + \|g\|_{L^4(\Omega\times(0,T);\gamma(\mathbb{R}^K,H_4))}\right). \tag{2.5}$$

The constant depends on D, ν, δ, the fixed finite smooth family b, and T_0, and is uniform in $T \leq T_0$. The solution map is causal and the same bound holds on common stopping intervals.

The additive stochastic theorem that we use is [1, Theorem 3.14]: if X_0 is isomorphic to a closed subspace of L^q, $q \geq 2$, and a shift of A has bounded H^∞ calculus of angle less than $\pi/2$, then $(A, 0)$ has stochastic maximal L^p regularity for $p > 2$ and the permitted weights. Here $p = 4$, the weight is zero, $X_0 = Y_4$, $\mathrm{D}(A) = V_4$, and the noise space is $[Y_4, V_4]_{1/2} = H_4$. The trace estimate follows from [1, Proposition 2.1, Definitions 3.7–3.8], using temporal smoothness $3/8 \in (1/4, 1/2)$. More explicitly, the estimate supplies

$$L^4(0,T_0;V_4)\cap H^{3/8,4}(0,T_0;[Y_4,V_4]_{5/8}) \hookrightarrow C\big([0,T_0];([Y_4,V_4]_{5/8},V_4)_{1/3,4}\big).$$

The trace space on the right is $(Y_4, V_4)_{3/4,4} = \mathcal{E}$, since $5/8 + (1/3)(1 - 5/8) = 3/4$. This gives the $C\mathcal{E}$ norm as well as the L^4V_4 norm. We always obtain these estimates on a fixed interval $[0, T_0]$ and then restrict them, so their constants do not come from a short-interval trace embedding. This is an additive theorem; the non-small transport in (2.4) requires the arguments below.

3 Flat transport models

At a boundary point a tangent vector field freezes to a horizontal constant vector. The corresponding model is a fixed periodic channel. Interior points require a three-dimensional torus, since the frozen vectors there may span all of $\mathbb{R}^3$. We first prove the channel estimate, retaining its pressure equation.

3.1 Channel resolvent and the colliding roots

Let $O = (\mathbb{R}^2/\Gamma)\times(0,h)$, where Γ is a fixed full-rank lattice. Write $H_q(O) = L^q_\sigma(O)$, $V_q(O) = W^{1,q}_{0,\sigma}(O)$, and $Y_q(O) = (V_{q'}(O))^*$, with periodic horizontal directions and zero vector trace at both walls in $V_q(O)$. Its horizontal dual frequencies satisfy $\xi\cdot\gamma \in 2\pi\mathbb{Z}$. Let

$$Q = \mathrm{diag}(Q_\tau, \nu), \qquad \delta I \leq Q_\tau \leq \nu I, \qquad A_Q = -\mathbb{P}_O\,\mathrm{div}(Q\nabla).$$

The projection and the pressure gradient are the ordinary ones. One could normalize diffusion by $x = Ly$, $L = \operatorname{diag}(Q_\tau^{1/2}, \sqrt{\nu})$, while leaving velocity components unchanged. Then both gradient and divergence contain $L^{-1}\nabla_y$, and the lattice becomes $Q_\tau^{-1/2}\Gamma$. Thus this normalization cannot remove the two horizontal weights while retaining the original projection. We work in the original coordinates.

For $\xi \neq 0$ put

$$k = |\xi|, \quad d_\xi = \xi^T Q_\tau \xi, \quad \alpha = \sqrt{(\lambda + d_\xi)/\nu}, \quad \operatorname{Re}\alpha > 0, \quad a = |\alpha|. \tag{3.1}$$

On $|\arg\lambda| \le \pi - \varepsilon$ we have $a \ge ck$, $\operatorname{Re}\alpha \ge ca$, and $|\lambda| + k^2 \asymp a^2$. Constants may depend on the fixed $\varepsilon, \delta, \nu, h, \Gamma$. The nonzero lattice has $k \ge k_0 > 0$. For velocity (v, w) and forcing (f_τ, f_3) the mode equations are

$$\begin{aligned} (\lambda + d_\xi - \nu\partial_z^2)v + i\xi p = f_\tau, \qquad & (\lambda + d_\xi - \nu\partial_z^2)w + p' = f_3, \\ i\xi \cdot v + w' = 0, \qquad & (v, w)|_{0,h} = 0. \end{aligned} \tag{3.2}$$

The transverse component is a scalar Dirichlet resolvent. Eliminating pressure in the remaining components gives

$$\nu(\partial_z^2 - k^2)(\partial_z^2 - \alpha^2)w = k^2 f_3 + i\xi \cdot f_\tau', \qquad w = w' = 0 \quad (z = 0, h), \tag{3.3}$$

with $v_\parallel = i\xi w'/k^2$ and $p = -((\lambda + d_\xi - \nu\partial_z^2)w' + i\xi \cdot f_\tau)/k^2$. At $\xi = 0$ the equation is instead $w = 0$, $p' = f_3$, and the tangential scalar Dirichlet problem. This mode has a spectral gap.

The roots α and k collide at $\lambda = \nu k^2 - d_\xi$, which can be positive. Put $\Delta = \alpha^2 - k^2$ and use

$$\phi_3(z) = \frac{\cosh(\alpha z) - \cosh(kz)}{\Delta}, \qquad \phi_4(z) = \frac{\sinh(\alpha z)/\alpha - \sinh(kz)/k}{\Delta}.$$

The removable values are understood analytically. After the two bottom conditions the remaining boundary matrix is

$$M = \begin{pmatrix} \phi_3(h) & \phi_4(h) \\ \phi_3'(h) & \phi_3(h) \end{pmatrix}, \quad \det M = \frac{(\alpha/k + k/\alpha)\sinh(\alpha h)\sinh(kh) - 2(\cosh(\alpha h)\cosh(kh) - 1)}{\Delta^2}.$$

At collision,

$$\det M = \frac{\sinh^2(kh) - (kh)^2}{4k^4} > 0. \tag{3.4}$$

Indeed the three entries are respectively $h\sinh(kh)/(2k)$, $(kh\cosh(kh) - \sinh(kh))/(2k^3)$, and $(\sinh(kh) + kh\cosh(kh))/(2k)$. The limiting determinant as $k \to 0$ is $h^4/12$; the actual zero mode was treated separately. For all other parameters, the homogeneous energy identity

$$\lambda \|u\|_2^2 + d_\xi \|u\|_2^2 + \nu \|u'\|_2^2 = 0$$

excludes a nonzero solution in the stated sector. Multiplication by $e^{-i\arg(\lambda)/2}$ also gives the coercive forced estimate.

For the kernel estimates use the divided difference

$$D_\alpha(r) = \frac{e^{-\alpha r} - e^{-kr}}{\alpha - k}, \qquad D_k(r) = -re^{-kr}, \qquad \Pi = \xi\xi^T/k^2.$$

The decaying half-channel Poisson matrix is

$$H_+(r) = \begin{pmatrix} e^{-\alpha r} I_2 + k D_\alpha(r)\Pi & (i\xi/k)\alpha D_\alpha(r) \\ i\xi^T D_\alpha(r) & e^{-kr} - k D_\alpha(r) \end{pmatrix}. \tag{3.5}$$

It equals I_3 at zero and satisfies (3.2) homogeneously. Let $J = \mathrm{diag}(1, 1, -1)$ and define

$$R_+ = H_+(h), \quad R_- = JH_+(h)J, \quad L = \begin{pmatrix} I & R_- \\ R_+ & I \end{pmatrix}, \quad B = L^{-1}.$$

Thus prescribed boundary values are obtained from $u(z) = H_+(z)c + JH_+(h-z)Je$ by solving $L(c, e)^T = (g_0, g_h)^T$.

Both B and $\mathrm{diag}(W, W)B\,\mathrm{diag}(W, W)^{-1}$ are uniformly bounded, where $W = \mathrm{diag}(I_2, a/k)$. To see this, for comparable a, k use coercivity and compactness. For $k \to \infty$ both versions of $R_\pm$ tend to zero. For bounded k and $a \to \infty$, putting $\vartheta = a/\alpha$,

$$R_+ \to e^{-kh} \begin{pmatrix} 0 & -i\widehat{\xi} \\ 0 & 1 \end{pmatrix}, \qquad W R_+ W^{-1} \to e^{-kh} \begin{pmatrix} 0 & 0 \\ -i\vartheta\widehat{\xi}^T & 1 \end{pmatrix}.$$

The reflected limits use J. The only possible nonzero eigenvalue of their product is $e^{-2kh} < 1$, uniformly away from one for $k \geq k_0$. On the remaining compact parameter set, the homogeneous boundary-value problem has no nonzero solution, so its boundary matrix is invertible. Consequently every boundary block of B has size

$$B_{ee'} = \begin{pmatrix} O(1) & O(1) \\ O(k/a) & O(1) \end{pmatrix}. \tag{3.6}$$

The lower-left bound uses the weighted inverse, the upper-right bound the unweighted inverse. A regular normalized determinant, useful also at collision, is

$$(1 - e^{-(\alpha+k)h})^2 - (\alpha + k)^2 \left(\frac{e^{-kh} - e^{-\alpha h}}{\alpha - k} \right)^2.$$

Its collision value is $4e^{-2kh}(\sinh^2(kh) - (kh)^2) > 0$.

3.2 Kernel derivatives and R-bounds

Here all horizontal derivatives keep λ fixed. For $c = Q_\tau \xi$,

$$k_i = \xi_i/k, \quad k_{ij} = \delta_{ij}/k - \xi_i\xi_j/k^3, \quad \alpha_i = \frac{c_i}{\nu\alpha}, \quad \alpha_{ij} = \frac{Q_{ij}}{\nu\alpha} - \frac{c_i c_j}{\nu^2\alpha^3}. \tag{3.7}$$

In particular $k^m \partial^\beta \alpha = O(k^2/a)$ for $m = |\beta| = 1, 2$. If $E_t(r) = e^{-tr}$, then

$$\begin{aligned} \partial_i(t^m E_t) &= E_t(mt^{m-1} - rt^m)t_i, \\ \partial_{ij}(t^m E_t) &= E_t[(m(m-1)t^{m-2} - 2mrt^{m-1} + r^2 t^m)t_i t_j + (mt^{m-1} - rt^m)t_{ij}]. \end{aligned} \tag{3.8}$$

This also holds for $m = -1$. The angular derivatives are, with $N = \xi\xi^T$,

$$\begin{aligned} \Pi_i &= (e_i\xi^T + \xi e_i^T)/k^2 - 2\xi_i N/k^4, \\ \Pi_{ij} &= (e_i e_j^T + e_j e_i^T)/k^2 - 2\{\xi_j(e_i\xi^T + \xi e_i^T) + \xi_i(e_j\xi^T + \xi e_j^T) + \delta_{ij}N\}/k^4 + 8\xi_i\xi_j N/k^6. \end{aligned} \tag{3.9}$$

For comparable roots put $t_s = (1-s)k + s\alpha$. Then

$$D_\alpha = -r\int_0^1 e^{-t_s r}\,ds, \qquad (D_\alpha)_i = r^2\int_0^1 e^{-t_s r}(t_s)_i\,ds, \tag{3.10}$$
$$(D_\alpha)_{ij} = r^2\int_0^1 e^{-t_s r}(t_s)_{ij}\,ds - r^3\int_0^1 e^{-t_s r}(t_s)_i(t_s)_j\,ds.$$

For $a \ge 2k$ the denominator $|\alpha - k| \ge a/2$ is safe: if $A = e^{-\alpha r} - e^{-kr}$, $C = \alpha - k$, then $D_i = (A_i - DC_i)/C$ and $D_{ij} = (A_{ij} - DC_{ij} - D_iC_j - D_jC_i)/C$. These formulas, with ordinary product differentiation, give all pure and mixed second derivatives in (3.5).

Let primes now denote distributional derivatives in z, and put

$$K_\alpha(r) = e^{-\alpha|r|}/(2\alpha), \qquad G(r) = \frac{e^{-k|r|}/(2k) - e^{-\alpha|r|}/(2\alpha)}{\Delta}.$$

The full-line particular velocity kernel is

$$R^0(r) = \nu^{-1}\begin{pmatrix} K_\alpha(I_2 - \Pi) - G''\Pi & i\xi G' \\ i\xi^T G' & k^2 G \end{pmatrix}. \tag{3.11}$$

Its crucial distributional identities are

$$G'''' = \delta_0 + \frac{k^3 e^{-k|r|} - \alpha^3 e^{-\alpha|r|}}{2\Delta}, \qquad K_\alpha'' = \alpha^2 K_\alpha - \delta_0. \tag{3.12}$$

Thus $\partial_z^2 R^0$ contains exactly $-\nu^{-1}\,\mathrm{diag}(I_2, 0)\delta_0$. The derivatives of its two angular projections cancel. Away from zero, $G^{(j)}$ is the signed version of $(-1)^j(k^{j-1}e^{-k|r|} - \alpha^{j-1}e^{-\alpha|r|})/(2\Delta)$, with the parity sign on crossing zero.

For $a \ge 2k$ use $|\Delta| \ge 3a^2/4$ and (3.7)–(3.8). Derivatives of Δ^{-1} are $-\Delta_i/\Delta^2$ and $2\Delta_i\Delta_j/\Delta^3 - \Delta_{ij}/\Delta^2$, where $\Delta_i = 2c_i/\nu - 2\xi_i$ and $\Delta_{ij} = 2Q_{ij}/\nu - 2\delta_{ij}$. For comparable roots use the nonsingular expression

$$G(r) = \frac{1}{2(\alpha + k)}\int_0^1 e^{-t_s|r|}\left(\frac{|r|}{t_s} + \frac{1}{t_s^2}\right)ds.$$

Since $\mathrm{Re}\, t_s \ge ck$, its derivatives of vertical order $j \le 3$ are bounded by $Ck^{j-3}(1 + k|r|)^M e^{-ck|r|}$; order four uses (3.12). In both regimes the kernels of

$$k^{|\beta|}\partial_\xi^\beta\{\lambda R^0,\ k^2 R^0,\ k\partial_z R^0,\ \partial_z^2 R^0\}, \qquad |\beta| \le 2, \tag{3.13}$$

are bounded by $Cke^{-ck|r|} + Cae^{-ca|r|}$, together with the identity term just described. Polynomial factors are absorbed by reducing c.

For the boundary correction set

$$P(z) = [H_+(z), JH_+(h - z)J], \qquad T(s) = \mathrm{col}(R^0(-s), R^0(h - s)).$$

The complete channel resolvent is

$$R(\lambda, \xi; z, s) = R^0(z - s) - P(z)BT(s). \tag{3.14}$$

Put $\mu = |\lambda| + k^2$. At either endpoint, with $d_e(s)$ its distance, the tangential and normal rows of the trace kernel obey

$$|k^{|\beta|}\partial^\beta T_\tau(s)| \le C\mu^{-1}(ae^{-cad_e(s)} + ke^{-ckd_e(s)}), \tag{3.15}$$
$$|k^{|\beta|}\partial^\beta T_3(s)| \le C\mu^{-1}k(e^{-cad_e(s)} + e^{-ckd_e(s)}), \quad |\beta| \le 2.$$

These follow separately from k^2G, kG', G'' in (3.11). For $a \geq 2k$, the Poisson blocks have sizes

$$\begin{aligned} H_{\tau\tau} &= O(E_a + (k/a)E_k), \qquad H_{\tau 3} = O(E_a + E_k), \\ H_{3\tau} &= O((k/a)(E_a + E_k)), \quad H_{33} = O(E_k + (k/a)E_a), \end{aligned} \tag{3.16}$$

where $E_b = e^{-cbr}$. They retain these sizes under normalized horizontal derivatives. In particular the slow response to a tangential input has the factor k/a.

For completeness, inverse differentiation gives

$$B_i = -BL_iB, \qquad B_{ij} = BL_jBL_iB + BL_iBL_jB - BL_{ij}B.$$

Differentiating WLW^{-1} and using $k\partial_i W\, W^{-1}$ and its second analogues, which are bounded by (3.7), proves (3.6) for all the normalized first and second horizontal derivatives of B. The second derivative of the reflected product has all nine terms

$$\begin{aligned} (PBT)_{ij} = {} & P_{ij}BT + P_iB_jT + P_iBT_j + P_jB_iT + PB_{ij}T \\ & + PB_iT_j + P_jBT_i + PB_jT_i + PBT_{ij}. \end{aligned} \tag{3.17}$$

For vertical order $j_0 = 0, 1, 2$ replace P by $\partial_z^{j_0}P$; multiply by λ or by k^{2-j_0} as appropriate in (3.13). Fast parts gain a^{j_0}, slow parts k^{j_0}. The fast factor $k^{2-j_0}a^{j_0} \leq Ca^2$ cancels μ^{-1}. Whenever an a-sized tangential trace enters a slow output, (3.16) and (3.6) supply k/a. For example, a fast tangential trace entering a slow tangential output in λPBT is bounded by

$$C|\lambda|\frac{k}{a}\frac{a}{\mu}e^{-ckd_e(z)-cad_{e'}(s)} \leq Cke^{-c'k(d_e(z)+d_{e'}(s))} \leq \frac{C}{d_e(z) + d_{e'}(s)}.$$

Here $|\lambda|/\mu \leq 1$ and $a \geq ck$. For the second spatial families the slow output contributes k^2, giving $k^3/\mu \leq Ck$ instead. Fast–fast terms are bounded by $Cae^{-ca(d_e(z)+d_{e'}(s))}$; a fast output with a slow input has size at most Ck with decay at least at scale k on both sides. The block bounds for B preserve these estimates, and the smaller normal row in (3.15) handles the remaining columns. The same reasoning applies to the normalized derivatives just listed. Every resulting boundary kernel is bounded by

$$C \sum_{e,e' \in \{0,h\}} \frac{1}{d_e(z) + d_{e'}(s)}. \tag{3.18}$$

For comparable roots $a \asymp k$, (3.10) gives polynomials in $kd_e(z)$ and $kd_{e'}(s)$ times the same exponentials; absorbing the polynomials by a smaller exponential constant gives the same estimate. This accounts for pure and mixed jets, the reflected product, and the distributional identity; individual mode energy bounds alone would not suffice.

Each positive kernel in (3.18) is bounded on $L^q(0,h)$: split $1/(z+s)$ at $s = z$ into the two Hardy operators, and reflect for the other endpoints. Minkowski's inequality in ℓ^2 therefore gives the same bound for diagonal families. The full-line exponential kernels are dominated by the one-dimensional Hardy–Littlewood maximal operator. Its vector-valued estimate [5, Theorem 1], with outer exponent q and sequence exponent two, gives the corresponding ℓ^2 bounds. These arguments work on $L^q(0,h;\ell^2_N)$ uniformly in N and with another Hilbert index. Recall that a family $\mathcal{T} \subset \mathcal{L}(X)$ is R-bounded if, for independent symmetric signs r_ℓ,

$$\left(\mathbb{E}_r \left\| \sum_\ell r_\ell T_\ell x_\ell \right\|_X^2\right)^{1/2} \leq C \left(\mathbb{E}_r \left\| \sum_\ell r_\ell x_\ell \right\|_X^2\right)^{1/2} \qquad (T_\ell \in \mathcal{T}).$$

On L^q, Khintchine's inequality identifies these randomized norms with the corresponding ℓ^2 square-function norms. The preceding bounds therefore show that the families in (3.13), now for R, are R-bounded, including arbitrary finite collections of λ's.

3.3 Domains, multipliers, and stochastic estimates

The multiplier theorem [6, Corollary 4.4] applies to UMD spaces of Fourier type $r > 1$ when all normalized total derivatives through $\lfloor n/r \rfloor + 1$ are R-bounded. The spaces $L^q(0,h;\mathbb{C}^3 \otimes \ell^2_N)$ have Fourier type $\min(q,q')$ and UMD constants uniform in N, by the Rademacher embedding into L^q; see [6, Remarks 2.3, 2.6, 2.10]. In horizontal dimension two, second derivatives suffice for every fixed finite $q > 1$. Apply the theorem to the diagonal symbol with entries $\lambda_j R(\lambda_j, \xi)$. The preceding estimates have the additional Hilbert-index uniformity required to do this. We obtain the R-resolvent bound on $L^q(\mathbb{R}^2 \times (0,h))$, and the spatial domain bounds from the other three families.

Here is the periodic restriction needed for this application. First take the standard lattice; the fixed linear change for general Γ is addressed below. Cut off the symbol smoothly below $k_0/2$, with value one at every nonzero lattice frequency. For a finite trigonometric polynomial $F(x) = \sum_j e^{ij\cdot x} x_j$, multiply by $\phi(\epsilon x)$, where $\widehat{\phi}$ is smooth and compactly supported. For each frequency j, after $y = \epsilon x$, the Fourier error is $[m(j+\epsilon\xi) - m(j)]\widehat{\phi}(\xi)$. It tends uniformly to zero with its first two derivatives. Two integrations by parts bound its inverse transform by $o(1)(1+|y|)^{-2}$, whose qth power is integrable in $\mathbb{R}^2$. Thus the error in the original L^q norm is $o(\epsilon^{-2/q})$. Periodic averaging transfers the Euclidean multiplier bound to the samples. The constants are uniform in the Hilbert index; density finishes the restriction. A fixed lattice change uses the pure as well as mixed second derivatives already proved. Add the separately solved scalar zero mode.

At $\lambda = 1$ the spatial bounds imply that the constructed resolvent maps full L^q forcing into $\mathrm{D}_q = W^{2,q}(O;\mathbb{C}^3) \cap V_q(O)$. Its pressure gradient follows from (3.2). Conversely $f = (1 + A_Q)u$ for $u \in \mathrm{D}_q$ has this solution by mode uniqueness. It is the ordinary projected resolvent on solenoidal forcing, so A_Q is closed, densely defined, and has exactly this domain. The zero-mode gap and the nonzero-mode bounds give invertibility at zero. Resolvent sectors give a causal analytic semigroup. For the time symbol $M(t) = A_Q(it + A_Q)^{-1}$, both M and $tM' = -itA_Q(it + A_Q)^{-2}$ are R-bounded. The dimension-one case of [6, Corollary 4.4] gives deterministic maximal L^p regularity, $1 < p < \infty$. Semigroup causality and zero extension of forcing give constants uniform in $T \le T_0$. In particular this holds at $q = 4, 4/3, 8, 8/7$.

We next need the isotropic channel calculus; a bounded Euclidean-domain Stokes theorem does not directly have this geometry. Reflect tangential components evenly and normal components oddly into the doubled torus, apply its Helmholtz projection, and restrict. The result fixes solenoidal zero-normal fields and annihilates gradients, hence is $\mathbb{P}_O$. Fourier multipliers give its L^q bounds and consistent adjoints, and reflection gives $H^{s,2}$ bounds for $0 < s < 1/2$. Let A_0 be the isotropic channel Stokes operator. The elliptic projection

$$\widetilde{Q} = A_0^{-1}\mathbb{P}_O(-\Delta_D)$$

is bounded on full L^q: its adjoint is $(-\Delta_D)A_{0,q'}^{-1}\mathbb{P}_O$, bounded by the strong-domain estimate. It fixes H_q and maps full-vector $W^{2,q} \cap W_0^{1,q}$ into D_q. Interpolating this retraction gives $[H_q, \mathrm{D}_q]_{1/2} = V_q$. The componentwise Dirichlet Laplacian is the odd part of the doubled-torus Laplacian and has its scalar H^∞ calculus.

We spell out the comparison criterion [9, Theorem 8.2]. Let (X_0, X_1) be a reflexive B-convex interpolation couple, $Y_j \subset X_j$ complemented by compatible projections P_j, and let $\mathsf{A}_j, \mathsf{B}_j$ be consistent sectorial families on these spaces. Assume that A_j has H^∞ calculus and B_j is almost R-sectorial. If $\alpha < 0 < \beta$ and

$$P_0(\dot{X}_{0,\alpha,\mathsf{A}_0}) = \dot{Y}_{0,\alpha,\mathsf{B}_0}, \qquad P_1(\dot{X}_{1,\beta,\mathsf{A}_1}) = \dot{Y}_{1,\beta,\mathsf{B}_1},$$

then B has H^∞ calculus on $[Y_0, Y_1]_t$ for $0 < t < 1$. The dots denote the homogeneous fractional scales of the specified operators.

As in [9, Propositions 9.14–9.16, Theorem 9.17], apply this to inverse operators. Put $p_0 = 2$, take $p_1 = 8$ or $8/7$, and set

$$X_j = L^{p_j}(O;\mathbb{C}^3), \quad Y_j = H_{p_j}(O), \quad P_j = \mathbb{P}_O, \qquad \mathsf{A}_j = (-\Delta_{\mathrm{Dir},p_j})^{-1}, \quad \mathsf{B}_j = A_{0,p_j}^{-1}.$$

Here $-\Delta_{\mathrm{Dir},p_j}$ is the componentwise Dirichlet Laplacian. Choose $\alpha = -s$, $0 < s < 1/4$, and $\beta = 1$. Inversion exchanges the fractional signs. Thus the first image condition reads

$$\mathbb{P}_O\,\mathrm{D}((-\Delta_{\mathrm{Dir},2})^s) = \mathbb{P}_O H^{2s,2} = H^{2s,2}_\sigma = \mathrm{D}(A^s_{0,2}).$$

The retraction and subtrace reflection bounds above give this identity. For the second, write

$$Z_{\Delta,p_1} = (W^{2,p_1'} \cap W_0^{1,p_1'})^*, \qquad Z_{A,p_1} = (\mathrm{D}_{p_1'})^*.$$

It is precisely $\mathbb{P}_O Z_{\Delta,p_1} = Z_{A,p_1}$. The strong-domain inclusion $\mathrm{D}_{p_1'} \subset W^{2,p_1'} \cap W_0^{1,p_1'}$ with its graph norm, followed by duality and Hahn–Banach extension, gives this surjection consistently with the ordinary projection. The scalar calculus was proved above; the explicit resolvent bounds give R-sectoriality of A_0 and its inverse and consistency across these exponents. All spaces in the couple are reflexive and B-convex. The criterion at $t = 2/3$ therefore gives calculus on H_4 from $p_1 = 8$, and on $H_{4/3}$ from $p_1 = 8/7$. Calculus for the inverse is equivalent to calculus for A_0 by $F(z) \mapsto F(1/z)$. This proves existence of isotropic H^∞ calculus. Its angle is less than $\pi/2$ by the already proved R-sectorial bounds and the angle comparison in [9, Theorem 4.1, Corollary 4.4]. This angle comparison is used only after calculus existence has been established.

The resulting negative-half scale identifies $Y_q = (V_{q'})^*$ and gives $[Y_q, V_q]_{1/2} = H_q$. For general Q, the same causal zero-initial solution map has the consistent endpoint bounds, for $1 < p < \infty$,

$$\begin{aligned} L^p(0,T;H_q) &\longrightarrow W^{1,p}(0,T;H_q) \cap L^p(0,T;\mathrm{D}_q), \\ L^p(0,T;Z_q) &\longrightarrow W^{1,p}(0,T;Z_q) \cap L^p(0,T;H_q), \qquad Z_q = (\mathrm{D}_{q'})^*. \end{aligned}$$

The second line is the one-step extrapolation, equivalently obtained by duality from the adjoint strong estimate. The isotropic retraction identities give

$$[Z_q, H_q]_{1/2} = Y_q, \qquad [H_q, \mathrm{D}_q]_{1/2} = V_q.$$

Interpolate the solution and its time derivative at parameter $1/2$. This proves deterministic maximal regularity of the weak form with domain V_q on Y_q. Integration by parts identifies the form with $\int Q\nabla u : \nabla v$. No general-$Q$ root identity is required. The real trace is $(Y_4, V_4)_{3/4,4} = B^{1/2}_{4,4,0,\sigma}$, obtained by interpolating isotropic levels on either side of spatial smoothness $1/2$, both above the trace threshold $1/4$.

For this model set $\mathcal{E}_O = (Y_4(O), V_4(O))_{3/4,4}$ and $\mathcal{X}_T(O) = L^4(\Omega; C([0,T];\mathcal{E}_O) \cap L^4(0,T;V_4(O)))$. The additive stochastic theorem stated in Section 2 applies to the isotropic weak operator: Y_4 is similar to the closed subspace H_4 of L^4. For general Q, solve the zero-initial isotropic stochastic equation for z_g, then correct it with the zero-initial deterministic A_Q solution forced by $(A_{0,w} - A_{Q,w})z_g + f$. The form difference is bounded $V_4 \to Y_4$. For nonzero $h \in \mathcal{E}$, first extend by $z_h = e^{-tA_{0,w}}h$ and use the same deterministic correction on $(A_{0,w} - A_{Q,w})z_h$. The isotropic real trace estimate bounds z_h in $C\mathcal{E} \cap L^4 V_4$. Both corrections have this trace by deterministic maximal regularity. These maps are causal and preserve predictability. They prove the full additive channel estimate, including random $\mathcal{F}_0$-measurable initial data, on the required spaces.

3.4 Interior torus and removal of constant transport

On a fixed full three-dimensional torus put $H_q = L^q_\sigma$, $V_q = W^{1,q}_\sigma$, $Y_q = (V_{q'})^*$, and $\mathcal{E}_O = (Y_4, V_4)_{3/4,4} = B^{1/2}_{4,4,\sigma}$. These norms include constants. For arbitrary constant vectors $\beta_1, \dots, \beta_K \in \mathbb{R}^3$ set

$$T_k = \beta_k \cdot \nabla, \qquad Q = \nu I - \tfrac{1}{2} \sum_k \beta_k \otimes \beta_k \geq \delta I, \qquad A_Q = -\operatorname{div}(Q\nabla).$$

Here $\mathbb{P}(0) = I$, and $\mathbb{P}(\xi) = I - \xi\xi^T/|\xi|^2$ otherwise. Let $J_0 = (1 - \Delta)^{1/2}$. It is an isomorphism from H_q to Y_q, and J_0^{-1} sends V_q to $W^{2,q}_\sigma$. For bounded holomorphic F, Cauchy's estimates give all normalized frequency derivatives through order four of $F(1 + \xi^T Q \xi)$. Scalar multipliers, or the scalar case of [6, Corollary 4.4], give a bounded H^∞ calculus for $1 + A_Q$ on H_q, with domain $W^{2,q}_\sigma$. Similarity by J_0^{-1} gives its weak calculus with domain V_q on Y_q.

The three-dimensional restriction can be proved by the same window argument as above. This time differentiate the compact Fourier error through order four and apply $(1 - \Delta_\xi)^2$. The inverse transform is $o(1)(1 + |y|^2)^{-2}$, integrable to the qth power for $q = 4, 4/3$ in $\mathbb{R}^3$. Periodic averaging and density finish the proof for a fixed, possibly rotated lattice. The scalar multiplier is smooth at zero; the projection's constant mode is kept separately. The calculus constants are uniform for $\delta I \leq Q \leq \nu I$. The additive stochastic theorem therefore applies on Y_4. One fixed shift handles the mean and is removed by an exponential change of unknown on $[0, T_0]$.

On either flat geometry consider

$$du + Au\,dt = f\,dt + \sum_k (T_k u + g_k)\,dW^k, \qquad A = -\nu \mathbb{P}_O \Delta. \tag{3.19}$$

In the channel the β_k are horizontal; on the torus they are arbitrary. Translations $U_t v(x) = v(x + \sum_k \beta_k W^k(t))$ are isometries of all the spaces just defined and commute with A, T_k. Since $A_Q = A + \frac{1}{2}\sum_k T_k^2$, Itô's product rule gives

$$dv + A_Q v\,dt = \left(U_t^{-1} f - \sum_k T_k U_t^{-1} g_k \right) dt + \sum_k U_t^{-1} g_k\,dW^k, \qquad v = U_t^{-1} u. \tag{3.20}$$

The correction involving g is essential. Each $T_k : H_4 \to Y_4$ is bounded by integration by parts. Thus all the transformed data are in the additive data spaces. Smooth horizontal regularization in the channel, and Fourier regularization on the torus, commute with these operators. They prove (3.20) first for smooth approximants. The $\mathcal{X}_T$ and stochastic integral bounds then pass to the weak equation for rough data. This proves the flat transport estimate, uniqueness, and causality.

On the torus the mean satisfies exactly

$$\overline{u}(t) = \overline{h} + \int_0^t \overline{f}(s)\,ds + \sum_k \int_0^t \overline{g}_k(s)\,dW^k(s).$$

Its finite-dimensional estimate controls its contribution to the full $W^{1,4}$ norm. There is no restriction on the rank of the family β. For a common stopping interval, extend only the inhomogeneous data by zero to $[0, T_0]$ and restrict the causal solution. The multiplicative coefficients still define the full-interval equation; uniqueness identifies the restrictions. Constants are uniform in $T \leq T_0$ and, for bounded β, in all covariances between δI and νI.

4 Localization on the weak Stokes scale

A cutoff followed by Helmholtz projection need not preserve zero vector trace. To localize simultaneously on V_q and Y_q we instead use the isotropic Stokes energy form. All roots in this section belong to isotropic operators. For either model domain O we use $H_q(O), V_q(O), Y_q(O)$ as above, with no wall trace on the full torus, and write $\mathcal{E}_O = (Y_4(O), V_4(O))_{3/4,4}$.

4.1 Energy-form localizers

For a physical domain D and a model O write $M_D = 1 + A_D$, $M_O = 1 + A_{0,O}$, and use the same letters for their consistent extrapolations. On the torus the shift includes constants. Let E be a smooth cutoff and coordinate map on full vector fields. Suppose E, E^* are L^q bounded and preserve full-vector $W_0^{1,q}$ and $W^{2,q} \cap W_0^{1,q}$. The latter means zero value trace, without a condition on derivative traces. The actual charts below have these properties for each fixed radius. Define

$$\langle Cu, v\rangle = \int_O (\nabla(\mathsf{E}u) : \nabla v + (\mathsf{E}u)\cdot v), \qquad T = M_O^{-1/2} C M_D^{-1/2}. \tag{4.1}$$

The form is bounded $V_q(D) \to Y_q(O)$, so $T : H_q(D) \to H_q(O)$. For $u \in \mathrm{D}(A_D)$ its strong representative is $Cu = \mathbb{P}_O(1 - \Delta_O)\mathsf{E}u \in H_q(O)$. Since $M_D^{-1/2} : V_q \to \mathrm{D}(A_D)$, this proves $T : V_q(D) \to V_q(O)$. Similarly

$$C^* v = \mathbb{P}_D \mathsf{E}^* (1 - \Delta_O) v, \qquad v \in \mathrm{D}(A_{0,O}),$$

gives $T^* : V_{q'}(O) \to V_{q'}(D)$. Duality extends T consistently to $Y_q(D) \to Y_q(O)$. We use both $q = 4$ and $q = 4/3$.

Define the elliptic projection on full vector fields by

$$Q_O = M_O^{-1} \mathbb{P}_O (1 - \Delta_O), \qquad S = Q_O \mathsf{E}. \tag{4.2}$$

Its full L^q extension follows by duality from the strong Stokes estimate, as in Section 3.3. It fixes solenoidal fields and preserves zero-value-trace orders one and two. Equations (4.1)–(4.2) give

$$T = M_O^{1/2} S M_D^{-1/2}. \tag{4.3}$$

We will compare this map with ordinary Piola localization, rather than assume that the latter acts on the weak dual scale.

4.2 Geometry, divergence correction, and pressure

Choose a regular radius-r cover of D with bounded overlap, $N \le Cr^{-3}$, and partition $\sum_j \chi_j = 1$. The two model cells are fixed independently of r: a full three-dimensional torus for interior patches and a periodic channel for boundary patches. Interior charts are affine rotations. Boundary charts, centered on ∂D, are volume-preserving graph shears

$$F_j(y) = x_j + O_j(y', y_3 + \psi_j(y')), \quad O_j \in SO(3), \quad G_j = DF_j, \quad R_j = G_j^{-1}, \quad \mathsf{E}_j u = \chi_j R_j (u \circ F_j). \tag{4.4}$$

In the chart formula, χ_j denotes the pulled-back cutoff $\chi_j \circ F_j$. Cutting off the graph outside a Cr neighborhood gives $\|\psi_j\|_\infty \le Cr^2$, $\|\nabla\psi_j\|_\infty \le Cr$, $\|D^2\psi_j\|_\infty \le C$, and uniform bi-Lipschitz bounds. Also $|\partial^m \chi_j| \le C_m r^{-m}$. Smooth composition and multiplication show the value-trace and strong-domain properties used above for both E_j and its integral adjoint. The frozen vectors are

$\beta_{j,k} = O_j^T b_k(x_j)$. They are horizontal at a boundary center and arbitrary at an interior center. Orthogonal freezing preserves (2.3).

We need a negative-order pressure estimate to compare the physical and flat Laplacians. Put

$$\eta = 1/8, \qquad d = 1/16, \qquad \zeta = 3/16, \qquad \theta = 1 - d = 15/16. \tag{4.5}$$

These are fixed proof parameters. The value η lies strictly inside both pressure and subtrace windows, $2d = \eta$ matches spatial and fractional orders, and $\theta = 1 - d$ yields a lower-order gain ρ^{-d}. The margin $\eta < \zeta < 1/4$ permits multiplication at order $-\eta$ and leaves $1 - \zeta - 3/4 = 1/16 > 0$ after summing the cover; see Section 4.4. The ordinary Neumann gradient projection in [3, Theorem 4.38, Proposition 4.39] is bounded on $H^{s,q}$ for $-1 + 1/q < s < 1/q$ on smooth bounded domains. Take $s = \pm\eta$ and $q = 4, 4/3$, strictly inside these intervals. If $\nabla p_u = (I - \mathbb{P}_D)\Delta u$, fixing its mean on each component, the projection bound and the Nečas estimate [3, Lemma 2.12] give

$$\|p_u\|_{H^{1-\eta,q}/\mathbb{R}} + \|\nabla p_u\|_{H^{-\eta,q}} \le C \|\Delta u\|_{H^{-\eta,q}}. \tag{4.6}$$

The potential estimate extends from smooth fields by density. The cited result is used in its fixed preprint version. For $G = D$ or a model O, use the isotropic scale $X_{s,G} = \mathrm{D}(M_{G,q}^s)$ for $s \ge 0$, and $X_{-s,G} = (\mathrm{D}(M_{G,q'}^s))^*$ for $s > 0$, with the integral pairing extending the $H_q, H_{q'}$ duality. This is equivalently the completion in the norm $\left\| M_{G,q}^{-s} \cdot \right\|_{H_q}$. The isotropic fractional scale then gives

$$\mathrm{D}(M_D^\theta) = H^{2-\eta,q} \cap V_q, \qquad X_{-d,O} = \mathbb{P}_O H^{-\eta,q},$$

the latter consistently by duality from $\mathrm{D}(M_{O,q'}^d) = H_\sigma^{\eta,q'}$, below the trace thresholds. Canonical zero extension at order $-\eta$ is bounded by [3, Proposition 2.4]; its strict window contains our exponents. In what follows we extend $\partial_{il} u$ and ∇p_u themselves, rather than differentiate the zero extension of u.

Fix one convex Euclidean box in each model coordinate cell, the boundary box having its bottom on the wall. It contains every small cutoff of that type. The regularized supported divergence solver of [4, formula (3.13), Theorem 3.2, Corollary 3.4, Remark 3.5] uses one smoothing kernel on each box and satisfies

$$\operatorname{div} \mathcal{R}_B g = g - \vartheta \int g, \qquad \mathcal{R}_B : H_{\mathrm{supp}}^{s,q} \to H_{\mathrm{supp}}^{s+1,q}, \quad s = -1/8, 0, 1. \tag{4.7}$$

Here $\int \vartheta = 1$. The same operator acts consistently at all these exponents and orders. Its supported outputs extend into the model; at order one they have zero value trace, and at order two the supported extension also has zero derivative traces. We use the Euclidean result on these fixed boxes, not a presumed torus version.

For solenoidal f, the Piola identity gives $\operatorname{div} \mathsf{E}_j f = \nabla \chi_j \cdot R_j (f \circ F_j)$, of mean zero. This follows first from the weak zero-normal identity and then by density on $X_{-d,D}$, using the consistent distributional extensions of the same Piola map and divergence correction. Consequently, suppressing j,

$$\begin{aligned} (\mathbb{P}_O - Q_O)\mathsf{E} f &= (\mathbb{P}_O - Q_O)\mathcal{R}_B \operatorname{div} \mathsf{E} f, \\ (S - \mathsf{E}) f &= (Q_O - I)\mathcal{R}_B \operatorname{div} \mathsf{E} f. \end{aligned} \tag{4.8}$$

On the channel, even tangential and odd normal reflection, followed by full-torus multipliers, bounds $\mathbb{P}_O$ on the supported positive spaces here; zero vector trace makes reflection bounded through order one. On the torus use its Fourier projection. Thus, for fixed r, $S - \mathsf{E} : H_q \to W^{1,q}$ and $S - \mathsf{E} : V_q \to W^{2,q}$. It is compact at orders zero and one, respectively. For an $H^{-1/8,q}$ input, the same correction gains one order before $\mathbb{P}_O, Q_O$ are applied.

4.3 Two defect estimates and square roots

With $S = Q_O\mathsf{E}$ define $K = M_O S - S M_D$. For a strong-domain field the exact identity is

$$Ku = (\mathbb{P}_O - Q_O)\mathsf{E}M_D u + \mathbb{P}_O\big[(1-\Delta_y)\mathsf{E}u - \mathsf{E}(1-\Delta_x)u - \mathsf{E}\nabla p_u\big]. \tag{4.9}$$

Using (4.4), the principal second-order and pressure terms are

$$\mathbb{P}_O\left[-\chi R(GG^T - I)_{il}(\partial_{il}u)\circ F + \chi(G^T - R)(\nabla p_u)\circ F\right]. \tag{4.10}$$

The nonorthogonal pressure term is retained. Both coefficients have supremum Cr and Lipschitz norm C, including cutoff derivatives. Their C^ζ norms are therefore at most $Cr^{1-\zeta} = Cr^{13/16}$. Composition with the extended shear is bounded on $H^{-\eta,q}$, uniformly in r, by duality and interpolation of its $L^{q'}$ and $W^{1,q'}$ bounds. Multiplication by C^ζ functions is bounded at order $-\eta$: the heat-kernel formula for $[J_0^\eta, a]$ has near-zero integral $\int_0^1 t^{-1+(\zeta-\eta)/2}\,dt < \infty$, and duality gives the negative-order bound. Together with (4.6), (4.10) is bounded by $Cr^{13/16}$ both as $\mathrm{D}(M_D) \to H_q(O)$ and as $\mathrm{D}(M_D^\theta) \to X_{-d,O}$.

The other Laplacian terms are

$$-(\Delta(\chi R))u\circ F - 2\partial_a(\chi R)G_{ia}(\partial_i u)\circ F - \chi R(\Delta F_i)(\partial_i u)\circ F.$$

The pressure remainder is $(\nabla\chi)p_u \circ F$, since $\mathbb{P}_O$ annihilates the flat gradient. They map $\mathrm{D}(M_D^\theta) \to H_q(O)$ with bound C_r. The first term of (4.9) has the same bound by (4.8) applied to $M_D u$ at order $-\eta$.

For $M_{i,\rho} = M_i + \rho$, $\rho \geq 1$, we have proved both estimates

$$\left\|KM_{D,\rho}^{-1}\right\| + \left\|M_{O,\rho}^{-d} K M_{D,\rho}^{d-1}\right\| \leq Cr^{13/16} + C_r\rho^{-d}. \tag{4.11}$$

For the lower-order term the first norm gains ρ^{-d} from $M_D^\theta M_{D,\rho}^{-1}$, the second from its left factor. The principal terms are bounded on both fractional scales. Choose r first, then one $\rho(r)$ for all finitely many charts and both dual exponents, so that the right side is at most $a(r) = Cr^{13/16}$. The coefficient C in this rate is uniform in r.

We give the root comparison quantitatively, since smallness on H_q alone is insufficient for the trace reconstruction. Apply [9, Theorem 6.1, Lemma 6.2] to $\mathcal{A} = \mathrm{diag}(M_{D,\rho}, M_{O,\rho})$ and the lower off-diagonal perturbation zK. Its range condition is automatic for the shifted invertible diagonal. The two smallness conditions are exactly (4.11). For $|z|a(r) < c$, the resolvent series gives uniform H^∞ bounds, and the small strong relative bound gives uniform graph norms. Bounded imaginary powers and complex interpolation [9, Proposition 2.2] give uniform square-root norms on the shifted V spaces. Explicitly, for $G = D, O$ set

$$\|v\|_{V_{q,\rho}(G)} = \left\|M_{G,\rho}^{1/2} v\right\|_{H_q(G)} \asymp \|v\|_{V_q(G)} + \rho^{1/2}\,\|v\|_{H_q(G)}, \qquad \rho \geq 1.$$

The equivalence is uniform in ρ by the isotropic calculus.

Conjugation of $\mathcal{A}$ by the triangular matrix with lower entry $-zS$ gives the perturbation with lower entry zK. Its square root has lower entry zF_ρ, where

$$F_\rho = M_{O,\rho}^{1/2} S - S M_{D,\rho}^{1/2}, \qquad D_\rho = F_\rho M_{D,\rho}^{-1/2}.$$

Taking $|z|$ comparable to $a(r)^{-1}$ in the uniform root bound gives $\|D_\rho\|_{H_q \to H_q} \leq Ca(r)$. The exact identity

$$M_{O,\rho}^{1/2} D_\rho M_{D,\rho}^{-1/2} = KM_{D,\rho}^{-1} - D_\rho \tag{4.12}$$

also gives $Ca(r)$ on the shifted V_q norm. For $q = 4$ the shifted real interpolation norm satisfies, uniformly in ρ,

$$\|v\|_{(H_4,V_{4,\rho})_{1/2,4}} \asymp \|v\|_{\mathcal{E}} + \rho^{1/4} \|v\|_{H_4}.$$

This follows by splitting the K-functional integral at $\rho^{-1/2}$. Interpolation yields

$$\|D_\rho v\|_{\mathcal{E}_O} \le Ca(r) \|v\|_{\mathcal{E}} + C_r \|v\|_{H_4}, \tag{4.13}$$

and the analogous V_4 estimate with a lower H_4 term.

Let $T_\rho = M_{O,\rho}^{1/2} S M_{D,\rho}^{-1/2}$. Then $T - \mathsf{E} = D_\rho + \mathcal{K}_r$, with $\mathcal{K}_r = (T - T_\rho) + (S - \mathsf{E})$. To justify compactness at the positive levels, keep the fixed shift and write

$$\begin{aligned} T - T_\rho = {} & (M_O^{1/2} - M_{O,\rho}^{1/2}) S M_D^{-1/2} \\ & + M_{O,\rho}^{1/2} S (M_D^{-1/2} - M_{D,\rho}^{-1/2}). \end{aligned} \tag{4.14}$$

The first root difference equals $-\rho(M_O^{1/2} + M_{O,\rho}^{1/2})^{-1}$ and gains one spatial order. Since $S : V_q \to V_q$, its term maps H_q into $\mathrm{D}(M_O)$, compactly into $V_q(O)$. The inverse-root difference in the second term maps H_q to $\mathrm{D}(M_D^{3/2})$, compactly into $\mathrm{D}(M_D)$. The strong-domain bound on S then gives compactness $H_q(D) \to V_q(O)$ for that term too. Together with (4.8) this makes $\mathcal{K}_r$ compact on H_q, V_q and, by interpolation, on $\mathcal{E}$. These are statements for the fixed selected ρ, with no uniform compactness assertion as $\rho \to \infty$. The adjoint form argument in Section 4.1 gives the consistent Y_4 action.

4.4 Noise, drift, and reconstruction

Write $B_{D,k} = \mathbb{P}_D(b_k \cdot \nabla)$ and $B_{j,k} = \beta_{j,k} \cdot \nabla$. For a no-slip solenoidal field,

$$B_{D,k} u = (b_k \cdot \nabla) u - (I - \mathbb{P}_D)((u \cdot \nabla) b_k). \tag{4.15}$$

The two unprojected derivatives have the same divergence and zero normal component: this follows from $\operatorname{div} b_k = \operatorname{div} u = 0$, tangency of b_k, and the zero vector trace of u. Their difference is solenoidal. The second term in (4.15) is bounded on H_4; density gives the identity on V_4.

The exact commutator decomposition is

$$\begin{aligned} T_j B_{D,k} - B_{j,k} T_j = {} & (T_j - \mathsf{E}_j) B_{D,k} + (\mathsf{E}_j B_{D,k} - \beta_{j,k} \cdot \nabla \mathsf{E}_j) \\ & + \beta_{j,k} \cdot \nabla (\mathsf{E}_j - T_j). \end{aligned} \tag{4.16}$$

The full constant derivative is used on the unprojected $\mathsf{E}_j u$. The middle leading term is $\chi_j R_j[(b_k \circ F_j) - G_j \beta_{j,k}] \cdot (\nabla u) \circ F_j$, bounded by $Cr \|u\|_{V_4}$. Cutoff, chart, and pressure terms are bounded by $C_r \|u\|_{H_4}$. The first and last terms use the H_4 and V_4 bounds of $T_j - \mathsf{E}_j$, respectively. Their compact remainders are compact $V_4 \to H_4$, also after the displayed derivatives. Summing the finite noise index in its γ norm gives a $C(a(r) + r)$ leading bound, modulo compact $V_4 \to H_4$ maps and lower H_4 norms.

The drift has the exact normalized identity

$$M_j^{-1/2} (M_{j,w} T_j - T_j M_{D,w}) M_D^{-1/2} = K_j M_D^{-1}. \tag{4.17}$$

It is small $V_4 \to Y_4$, modulo a compact remainder: for the lower term use $M_D^{-1/2} : V_4 \to \mathrm{D}(M_D)$ followed by the compact inclusion into $\mathrm{D}(M_D^\theta)$. Identity shifts cancel.

The partition and bounded overlap give

$$\|u\|_{V_4} \le C\Big(\sum_j \|\mathsf{E}_j u\|_{W^{1,4}}^4\Big)^{1/4} + C_r \|u\|_{H_4}\,, \tag{4.18}$$
$$\|u\|_{\mathcal{E}} \le C\Big(\sum_j \|\mathsf{E}_j u\|_{B^{1/2}_{4,4}}^4\Big)^{1/4} + C_r \|u\|_{H_4}\,.$$

For the second inequality use the $W^{1/2,4} = B^{1/2}_{4,4}$ difference integral. Pairs at distance less than cr are reconstructed from a uniformly bounded family of enlarged patches. The charts are uniformly bi-Lipschitz and their Piola matrices uniformly Lipschitz. Pairs at distance at least cr cost only $C_r \|u\|_{L^4}^4$. This proves the uniform principal constant in (4.18).

Replacing E_j by T_j costs at most

$$CN^{1/4}a(r) \le Cr^{-3/4}r^{13/16} = Cr^{1/16} \longrightarrow 0. \tag{4.19}$$

The leading noise and drift errors have the same bound. Choose r small, then the common $\rho(r)$ of (4.11). There remain finitely many compact maps and lower H_4 norms. Ehrling's inequality, with the continuous embedding into L^2, bounds them by $\epsilon \|u\|_{V_4} + C_{r,\epsilon} \|u\|_{L^2}$, or at trace level by $\epsilon \|u\|_{\mathcal{E}} + C_{r,\epsilon} \|u\|_{L^2}$. Here $\mathcal{E}$ embeds compactly into H_4. Absorbing these terms proves both reconstruction inequalities with T_j and an L^2 remainder. For the continuous time norm use $\sup_t \sum_j a_j(t)^4 \le \sum_j \sup_t a_j(t)^4$; for the integral norm use Fubini. Thus the same reconstruction works in $\mathcal{X}_T$, including its actual continuous Besov trace.

5 The physical linear equation

We now prove Theorem 2.1. Scale the physical noise to sB, $0 \le s \le 1$. The covariances of every frozen model remain between δI and νI, and the vectors remain bounded. The channel matrix limits, kernel and multiplier estimates, and interior calculus are uniform on this compact parameter set. The isotropic transfers do not depend on s. Thus the radius and common shift in Section 4 can be fixed before choosing s.

For a solution already in $\mathcal{X}_T$, apply the flat-model estimate to $v_j = T_j u$. Its drift data are $T_j f + \nu(A_{j,w}T_j - T_j A_{D,w})u$, and its noise is

$$sB_j v_j + (T_j sB_D - sB_j T_j)u + T_j g.$$

The maps T_j act consistently on $Y_4, H_4, \mathcal{E}$ and preserve predictability. Using (4.16)–(4.19), then the compact-error absorption, gives

$$\|u\|_{\mathcal{X}_T} \le C_r\big(\|h\|_{L^4\mathcal{E}} + \|f\|_{L^4 Y_4} + \|g\|_{L^4\gamma(H_4)}\big) + C_r \|u\|_{L^4(\Omega;C([0,T];L^2))}\,. \tag{5.1}$$

The abbreviated data norms here and below are those of Theorem 2.1.

The remaining norm is controlled by physical energy. Orthogonality of $\mathbb{P}_D$ on L^2 and tangency and solenoidality of b_k give $(u, B_{D,k}u)_2 = 0$ and

$$2\nu \|\nabla u\|_2^2 - s^2 \sum_k \|B_{D,k}u\|_2^2 \ge 2\delta \|\nabla u\|_2^2\,. \tag{5.2}$$

On bounded D, $V_2 \hookrightarrow V_{4/3}$, so $Y_4 \hookrightarrow V_2^*$; also $H_4 \hookrightarrow H_2$. The variational Itô identity is therefore applicable with the triple V_2, H_2, V_2^*. It can equally be obtained by isotropic L^2 Stokes spectral

projections, using their uniform boundedness on V_2, V_2^* and the drift and noise L^2 bounds. The cross term with g in the quadratic variation is absorbed by $\delta \|\nabla u\|_2^2$. The homogeneous-noise martingale vanishes; the remaining martingale contains $(u, g_k)_2$. The stochastic maximal inequality, followed by Young's inequality, gives

$$\|u\|_{L^4(\Omega;CL^2)} \le C_{T_0} \left[\|h\|_{L^4(\Omega;L^2)} + T_0^{1/4} (\|f\|_{L^4 Y_4} + \|g\|_{L^4\gamma(H_4)}) \right]. \tag{5.3}$$

The time factor follows from $L^4(0,T) \hookrightarrow L^2(0,T)$. The estimate is uniform in s. Combining (5.1) and (5.3) proves (2.5) as an a priori estimate. The choices were r, then ρ, then the compact-error constants; none depends on the solution, $T \le T_0$, or s.

Existence follows by a finite continuation in s. At $s = 0$, the physical isotropic weak Stokes calculus and the additive theorem in Section 2 give a causal solution operator $\mathcal{S}_0$. If $\mathcal{S}_s$ exists, solve

$$u = \mathcal{S}_s(h, f, g + \Delta s\, B_D u)$$

by contraction in predictable $\mathcal{X}_T$. The constant is at most $C\Delta s \|B_D\|_{V_4 \to \gamma(\mathbb{R}^K, H_4)}$, with the uniform C just proved. Choose one positive step making this less than $1/2$; finitely many steps reach $s = 1$. The deterministic and stochastic integral estimates pass the limits to the equation in CY_4. Energy proves linear uniqueness. Each step preserves causality and restriction to common stopping times. This constructs the operator asserted in Theorem 2.1, rather than inferring existence from an a priori bound. □

6 The nonlinear equation

The remaining argument uses only the physical spaces in Section 2 and the constructed linear solution operator. Define the convection term on $\mathcal{E}$ by

$$\langle F(u), \varphi \rangle = \int_D (u \otimes u) : \nabla \varphi, \qquad \varphi \in V_{4/3}. \tag{6.1}$$

This is precisely $-\mathbb{P}_D \operatorname{div}(u \otimes u)$ in the weak formulation.

6.1 Product bound and adapted local construction

We use $\mathcal{E} \hookrightarrow L^8(D)$. A bounded extension of $W^{1/2,4}(D) = B^{1/2}_{4,4}(D)$ to $\mathbb{R}^3$ is obtained from a finite smooth atlas and even reflection. For cross-boundary pairs, reflection only decreases the distance between the two reflected points, so the fractional difference integral is bounded by the half-space integral. Smooth bi-Lipschitz changes and cutoffs preserve this bound. If $\Delta_j h$ is a dyadic decomposition of the extension, Bernstein and Hölder give

$$\|h\|_8 \le C \sum_{j \ge -1} 2^{-j/8} (2^{j/2} \|\Delta_j h\|_4) \le C \|h\|_{B^{1/2}_{4,4}},$$

since $\sum_{j \ge -1} 2^{-j/6} < \infty$. Consequently

$$\|F(u)\|_{Y_4} \le C \|u\|_{\mathcal{E}}^2, \qquad \|F(u) - F(v)\|_{Y_4} \le C(\|u\|_{\mathcal{E}} + \|v\|_{\mathcal{E}}) \|u - v\|_{\mathcal{E}}. \tag{6.2}$$

This avoids any assertion that $\mathbb{P}_D$ preserves zero vector trace on $W^{1,4}$.

Choose a Lipschitz $\chi : [0,\infty) \to [0,1]$, equal to one on $[0,1]$ and zero on $[2,\infty)$, and let $F_R(u) = \chi(\|u\|_{\mathcal{E}} / R) F(u)$. It is bounded by CR^2 and globally Lipschitz with constant $L_R \le CR$.

Indeed, if $\|u-v\|_{\mathcal{E}} < R$ and one cutoff is nonzero, both states have norm less than $3R$; expand the cutoff difference and use (6.2). If their distance is at least R, use the CR^2 bound. If both cutoffs vanish there is nothing to prove. For an adapted $\mathcal{E}$-continuous process this is a predictable state cutoff.

Let $\mathcal{S}_T$ denote the solution map in Theorem 2.1. In the closed Banach space of predictable $\mathcal{X}_T$ processes set $\Phi(u) = \mathcal{S}_T(h, F_R(u), 0)$. For deterministic $h, k \in \mathcal{E}$ its estimates give

$$\|\Phi(u) - \Phi(v)\|_{\mathcal{X}_T} \le C_L L_R T^{1/4} \|u - v\|_{\mathcal{X}_T}.$$

Choose

$$T_R = \min\{T_0, (2C_L \max(1, L_R))^{-4}\}. \tag{6.3}$$

Contraction yields a predictable $U_R^h \in \mathcal{X}_{T_R}$ and

$$\left\|U_R^h - U_R^k\right\|_{\mathcal{X}_{T_R}} \le 2C_L \|h - k\|_{\mathcal{E}}. \tag{6.4}$$

The integral estimates pass the fixed-point limit to the truncated equation in CY_4. For $R > \|h\|_{\mathcal{E}}$ set

$$\tau_R^h = \inf\{t \le T_R : \left\|U_R^h(t)\right\|_{\mathcal{E}} \ge R\} \wedge T_R, \qquad \inf \varnothing = \infty.$$

This stopping time is almost surely positive by continuity. On its closed interval the cutoff is one, including at norm R, so $u^h = U_R^h|_{[0,\tau_R^h]}$ solves the original equation. Its paths have the desired Besov continuity and $L^4 W_0^{1,4}$ regularity; no V_4 value at the exit time is required.

6.2 Uniqueness in the entire path class

Let u, v be any two adapted local solutions in $CB_{4,4}^{1/2} \cap L^4 W_0^{1,4}$, with the same initial datum and Brownian motions, on lifetimes τ_u, τ_v. They have $\mathcal{E}$-continuous paths: the divergence and trace constraints hold at almost every time from V_4, and then at every time by continuity and closedness. No moments are assumed for these paths. For deterministic T stop at $\tau_u \wedge \tau_v \wedge T$ and when the continuous running quantity

$$\|u(t)\|_{\mathcal{E}} + \|v(t)\|_{\mathcal{E}} + \int_0^t (\|u(s)\|_{V_4}^4 + \|v(s)\|_{V_4}^4)\, ds$$

first reaches N. Denote the resulting stopping time by σ_N. These times increase to the common lifetime capped by T.

On the stopped interval $w = u - v$ belongs to $CH_2 \cap L^2 V_2$. Convection lies in $L^2 V_2^*$, since both states are bounded in time in L^4, and $B_D w \in L^2 H_2$. The required second moments are finite after this norm stopping. Apply Itô's formula to finite isotropic L^2 Stokes spectral projections until σ_N. They converge on V_2, V_2^* and are uniformly bounded there; the drift pairings converge by the L^2 dual estimates, quadratic variations by $L^2 H_2$ convergence, and the martingales by the isometry and maximal inequality. The CH_2 bound permits uniform passage in time and then an almost sure subsequence. This proves the variational identity until σ_N, as also formulated in [1, Lemma A.5]. It does not extend the stopped state as a V_4-valued constant after σ_N.

By (5.2) the noise correction leaves $2\delta \|\nabla w\|_2^2$. Also $(w, B_{D,k} w)_2 = 0$. Convection by u is skew. Integrating the derivative off v in the remaining term and using Young's inequality gives, for $t \le \sigma_N$,

$$\|w(t)\|_2^2 + \delta \int_0^t \|\nabla w\|_2^2 \, ds \le \|w(0)\|_2^2 + \delta^{-1} \int_0^t \|v\|_\infty^2 \|w\|_2^2 \, ds. \tag{6.5}$$

The limiting martingale vanishes identically. Since $W^{1,4}(D) \hookrightarrow L^\infty(D)$ and $L^4(0,T) \hookrightarrow L^2(0,T)$, the Gronwall coefficient is finite pathwise. Equal initial data give $w = 0$ until σ_N. Let N and then T increase through integers; path continuity includes the endpoints. This proves uniqueness in the entire asserted class and identifies constructions with different cutoff radii on their common lifetimes.

6.3 Dependence after common stopping

Fix h and R with $\|h\|_{\mathcal{E}} < R/4$. If smooth admissible $h_n \to h$ in $\mathcal{E}$, the same bound holds eventually for h_n. Use one T_R, probability space, filtration, coefficient family and Brownian coupling to construct $U = U_R^h$ and $U_n = U_R^{h_n}$. Set

$$\begin{aligned} \tau &= \inf\{t \le T_R : \|U(t)\|_{\mathcal{E}} \ge R/2\} \wedge T_R, \\ \tau_n &= \inf\{t \le T_R : \|U_n(t)\|_{\mathcal{E}} \ge R\} \wedge T_R, \qquad \sigma_n = \tau \wedge \tau_n. \end{aligned} \tag{6.6}$$

These stopping times are almost surely positive. Both restrictions solve the original equation up to σ_n. For paths on this interval define

$$d_\sigma(u,v) = \sup_{t \le \sigma} \|u(t) - v(t)\|_{\mathcal{E}} + \left(\int_0^\sigma \|u - v\|_{V_4}^4 \, dt \right)^{1/4}. \tag{6.7}$$

Restriction of (6.4) and Markov's inequality give

$$\mathbb{P}\{d_{\sigma_n}(U_n, U) > \epsilon\} \le C\epsilon^{-4} \|h_n - h\|_{\mathcal{E}}^4 \longrightarrow 0. \tag{6.8}$$

If $\tau_n < \tau$, the perturbed path reached R while the base path had norm at most $R/2$. Its full-interval distance from U was therefore at least $R/2$, and

$$\mathbb{P}\{\sigma_n < \tau\} \le CR^{-4} \|h_n - h\|_{\mathcal{E}}^4 \longrightarrow 0. \tag{6.9}$$

The constants depend on the fixed coefficients, domain, T_0, R, and not on n. On the event $\sigma_n = \tau$, both original solutions reach the same base time τ. Equations (6.8)–(6.9) give continuity in probability in the full path topology. The perturbed solution reaches the base stopping time with probability tending to one. No positivity of an infinite infimum of exit times is needed. Together with whole-class uniqueness and the local construction, this proves the main result. □

Use of artificial intelligence. OpenAI Codex was used extensively to develop the argument, inspect primary sources, draft the manuscript, and perform internal mathematical and editorial checks. Two additional Codex contexts read a fixed draft, one as a reader and one as a referee, without the author checks or each other's report. These same-provider AI reviews and the author checks do not constitute human peer review or acceptance by the problem repository.

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